

1: Exercise 1.2.1 in the textbook

$$f(x, y) = 3x^2 + y^4, \nabla f(x, y) = (6x, 4y^3), \nabla^2 f(x, y) = \begin{bmatrix} 6 & 0 \\ 0 & 12y^2 \end{bmatrix}$$

(a) $\mathbf{x}_0 = (1, -2)$, $s = 1$, $\sigma = 0.1$, $\beta = 0.5$

We use the steepest descent algorithm and so $\mathbf{x}^1 = \mathbf{x}^0 + a^0 \mathbf{d}^0$, where $\mathbf{d}^0 = -\nabla f(\mathbf{x}_0)$. Moreover, we select the step size using the Armijo rule, i.e we set $a^0 = \beta^m s$ at the m th trial until the condition $f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) \leq \sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle$ is met. We have that $f(\mathbf{x}_0) = f(1, -2) = 19$, $\nabla f(\mathbf{x}_0) = \nabla f(1, -2) = (6, -32)$, $\mathbf{d}^0 = (-6, 32)$ and $\langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle = -1060$.

- $a^0 = s = 1$
 $f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) = 56 + 30^4$, $\sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle = -106$
 $\Rightarrow f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) > \sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle \Rightarrow \text{Armijo rule condition not satisfied}$
- $a^0 = \beta s = 0.5$
 $f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) = 14^4 - 7$, $\sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle = -53$
 $\Rightarrow f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) > \sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle \Rightarrow \text{Armijo rule condition not satisfied}$
- $a^0 = \beta^2 s = 0.25$
 $f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) = 1277.75$, $\sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle = -26.5$
 $\Rightarrow f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) > \sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle \Rightarrow \text{Armijo rule condition not satisfied}$
- $a^0 = \beta^3 s = 0.125$
 $f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) = -2.8125$, $\sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle = -13.25$
 $\Rightarrow f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) > \sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle \Rightarrow \text{Armijo rule condition not satisfied}$
- $a^0 = \beta^4 s = 0.0625$
 $f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) = -17.82$, $\sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle = -6.625$
 $\Rightarrow f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) \leq \sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle \Rightarrow \text{Armijo rule condition satisfied}$
Hence, the Armijo rule proposes $a^0 = 0.0625$. Thus, $\mathbf{x}^1 = \mathbf{x}^0 + a^0 \mathbf{d}^0 = (0.625, 0)$

(b) $\mathbf{x}_0 = (1, -2)$, $s = 1$, $\sigma = 0.1$, $\beta = 0.1$

- $a^0 = s = 1$
 $f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) = 75 + 30^4 - 19 = 56 + 30^4$, $\sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle = -106$
 $f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) \geq \sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle \Rightarrow \text{Armijo rule condition not satisfied}$
- $a^0 = \beta s = 0.1$
 $f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) = -16.4464$, $\sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle = -10.6$
 $f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) \leq \sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle \Rightarrow \text{Armijo rule condition satisfied}$

Hence, the Armijo rule proposes $a^0 = 0.1$. Thus, $\mathbf{x}^1 = \mathbf{x}^0 + a^0 \mathbf{d}^0 = (0.4, 1.2)$. In this case Armijo rule requires less trials than in the previous question.

(c) $\mathbf{x}_0 = (1, -2)$, $s = 1$, $\sigma = 0.1$, $\beta = 0.5$

We use the Newton method and so $\mathbf{x}^1 = \mathbf{x}^0 + a^0 \mathbf{d}^0$, where $\mathbf{d}^0 = -(\nabla^2 f(\mathbf{x}_0))^{-1} \nabla f(\mathbf{x}_0)$.

We have that $\mathbf{d}^0 = - \begin{bmatrix} 1/6 & 0 \\ 0 & 1/48 \end{bmatrix} \begin{bmatrix} 6 \\ -32 \end{bmatrix} = \begin{bmatrix} -1 \\ 2/3 \end{bmatrix}$ and $\langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle = -27.33$

• $a^0 = s = 1$

$$f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) = -15.84, \sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle = -2.73$$

$$f(\mathbf{x}^0 + a^0 \mathbf{d}^0) - f(\mathbf{x}^0) \leq \sigma a^0 \langle \nabla f(\mathbf{x}_0), \mathbf{d}^0 \rangle \Rightarrow \text{Armijo rule condition satisfied}$$

Hence, the Armijo rule proposes $a^0 = 1$. Thus, $\mathbf{x}^1 = \mathbf{x}^0 + a^0 \mathbf{d}^0 = (0, -4/3)$.

We notice that the cost of the new iterate is larger than the respective cost obtained in (a). Moreover, in this case it only took one trial of the Armijo rule in order to get the respective stepsize, however finding the new iterate involves finding the inverse of the Hessian.

1: Exercises 1.2.2, 1.2.3, 1.2.7

Look at the end of this pdf file for the handwritten notes.

2:

(a) If a second-order continuously differentiable function $f(x) : \mathbb{R} \rightarrow \mathbb{R}$ is convex, we have

$$f(\lambda y + (1 - \lambda)x) \leq \lambda f(y) + (1 - \lambda)f(x), \forall \lambda \in [0, 1], \forall x, y \in \mathbb{R}$$

After rewriting we have

$$f(x + \lambda(y - x)) \leq f(x) + \lambda(f(y) - f(x)), \forall \lambda \in [0, 1], \forall x, y \in \mathbb{R}$$

$$f(y) - f(x) \geq \frac{f(x + \lambda(y - x)) - f(x)}{\lambda}, \forall \lambda \in [0, 1], \forall x, y \in \mathbb{R}$$

As $\lambda \rightarrow 0$, we get

$$f(y) \geq f(x) + f'(x)(y - x), \forall x, y \in \mathbb{R},$$

$$f(x) \geq f(y) + f'(y)(x - y), \forall x, y \in \mathbb{R}.$$

This implies that

$$f'(y)(y - x) \geq f(y) - f(x) \geq f'(x)(y - x).$$

For $x \neq y$, we have

$$\frac{f'(y) - f'(x)}{y - x} \geq 0, \forall x \neq y \in \mathbb{R}.$$

It follows that

$$f''(x) = \lim_{y \rightarrow x} \frac{f'(y) - f'(x)}{y - x} \geq 0.$$

That is, the second-order derivative is non-negative.

(b) By the mean value version of Taylor's theorem we have (Prop. A.23 in the textbook)

$$f(y) = f(x) + f'(x)(y - x) + \frac{1}{2}f''(z)(y - x)^2, z \in [x, y]$$

If $f''(x) \geq 0, \forall x$, we have

$$f(y) \geq f(x) + f'(x)(y - x).$$

Let $z = \lambda x + (1 - \lambda)y$, we have

$$f(x) \geq f(z) + f'(z)(x - z),$$

$$f(y) \geq f(z) + f'(z)(y - z).$$

Multiplying by λ , $1 - \lambda$ and adding, we get

$$\lambda f(x) + (1 - \lambda)f(y) \geq f(z) + f'(z)(\lambda x + (1 - \lambda)y - z) = f(z) = f(\lambda x + (1 - \lambda)y)$$

So $f(x)$ is convex.

3:

(a) We know that f is convex. Then for $\lambda \in (0, 1]$, $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ it holds that

$$\begin{aligned} f((1 - \lambda)\mathbf{x} + \lambda\mathbf{y}) &\leq (1 - \lambda)f(\mathbf{x}) + \lambda f(\mathbf{y}) \Rightarrow \lambda f(\mathbf{y}) \geq \lambda f(\mathbf{x}) + f((1 - \lambda)\mathbf{x} + \lambda\mathbf{y}) - f(\mathbf{x}) \Rightarrow \\ f(\mathbf{y}) &\geq f(\mathbf{x}) + \frac{f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\lambda}. \end{aligned}$$

We set $\mathbf{h} = \mathbf{y} - \mathbf{x}$ and we compute the limit on both sides

$$\lim_{\lambda \rightarrow 0} f(\mathbf{y}) \geq \lim_{\lambda \rightarrow 0} \left\{ f(\mathbf{x}) + \frac{f(\mathbf{x} + \lambda\mathbf{h}) - f(\mathbf{x})}{\lambda} \right\} \Rightarrow f(\mathbf{y}) \geq f(\mathbf{x}) + f'(\mathbf{x}; \mathbf{h}),$$

where $f'(\mathbf{x}; \mathbf{h})$ is the directional derivative of f at $\mathbf{x}$ towards the direction $\mathbf{h}$. We also know from calculus that $f'(\mathbf{x}; \mathbf{h}) = \langle \nabla f(\mathbf{x}), \mathbf{h} \rangle$. Thus, $f(\mathbf{y}) - f(\mathbf{x}) \geq \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle$. Finally, by flipping $\mathbf{x}, \mathbf{y}$ we can get $f(\mathbf{x}) - f(\mathbf{y}) \geq \langle \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle, \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

(b) We know that f and g are convex function, thus it holds that

$$f(\mathbf{x}) - f(\mathbf{y}) \geq \langle \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n \quad (1)$$

and

$$g(x) - g(y) \geq g'(y)(x - y), x, y \in \mathbb{R}. \quad (2)$$

Also, using chain rule we can get

$$\nabla g(f(\mathbf{y})) = g'(f(\mathbf{y}))\nabla f(\mathbf{y}) \quad (3)$$

Then, combining (1), (2) and (3) and the the fact that g is non-decreasing, i.e $g'(x) \geq 0, \forall x$, we get that

$$\begin{aligned} g(f(\mathbf{x})) - g(f(\mathbf{y})) &\geq g'(f(\mathbf{y}))(f(\mathbf{x}) - f(\mathbf{y})) \geq g'(f(\mathbf{y}))\langle \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle = \langle \nabla g(f(\mathbf{y})), \mathbf{x} - \mathbf{y} \rangle \Rightarrow \\ (g \circ f)(\mathbf{x}) - (g \circ f)(\mathbf{y}) &\geq \langle \nabla(g \circ f)(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \end{aligned}$$

As a result, $g(f(\mathbf{x}))$ is a convex function.

(c) Let $g(y) = -y$. The function g is convex (because linear functions are convex and concave) and also decreasing. Then, $g(f(\mathbf{x})) = -f(\mathbf{x})$. By definition $f(\mathbf{x})$ is convex and thus $g(f(\mathbf{x})) = -f(\mathbf{x})$ is concave.

3:

(a) Let $f(\mathbf{x}, \mathbf{y})$ be the objective function. We prove that f is non-convex. Let $\mathbf{x}_1 = \mathbf{0}$ and $\mathbf{y}_1 = 10\mathbf{b}$ and $\mathbf{x}_2 = 10\mathbf{a}$ and $\mathbf{y}_2 = \mathbf{0}$. Then $f(\mathbf{x}_1, \mathbf{y}_1) = f(\mathbf{x}_2, \mathbf{y}_2) = \frac{1}{2}\|\mathbf{A}\|^2$, while

$$f\left(\frac{\mathbf{x}_1 + \mathbf{x}_2}{2}, \frac{\mathbf{y}_1 + \mathbf{y}_2}{2}\right) = \frac{1}{2}\|\mathbf{A} - 25\mathbf{A}\|^2 = 12\|\mathbf{A}\|^2 > \frac{1}{2}f(\mathbf{x}_1, \mathbf{y}_1) + \frac{1}{2}f(\mathbf{x}_2, \mathbf{y}_2)$$

since $\|\mathbf{A}\|^2 > 0$ by definition. Therefore, f is not convex.

(b) We have $f(\mathbf{x}, \mathbf{y}) = \frac{1}{2} \sum_{i,j=1}^n (a_i b_j - x_i y_j)^2$. Simple calculation shows that $\frac{\partial f}{\partial x_i} = x_i \|\mathbf{y}\|^2 - a_i < \mathbf{b}, \mathbf{y} >$ and $\frac{\partial f}{\partial y_j} = y_j \|\mathbf{x}\|^2 - b_j < \mathbf{a}, \mathbf{x} >$. To compute stationary points, we set

$$\frac{\partial f}{\partial x_i} = 0; \quad \frac{\partial f}{\partial y_j} = 0,$$

and follow the following case analysis.

If $\|\mathbf{x}\| = 0$, then $\frac{\partial f}{\partial y_j} = 0$ for any j . Since there exists i such that $a_i \neq 0$, $\frac{\partial f}{\partial x_i} = 0$ yields $< \mathbf{b}, \mathbf{y} > = 0$. Therefore, $\mathbf{x}, \mathbf{y}$ that satisfy $\|\mathbf{x}\| = 0$ and $< \mathbf{b}, \mathbf{y} > = 0$ are stationary points. Similarly, $\mathbf{x}, \mathbf{y}$ that satisfy $\|\mathbf{y}\| = 0$ and $< \mathbf{a}, \mathbf{x} > = 0$ are also stationary points.

Now we consider the case that $\|\mathbf{x}\| \neq 0$ and $\|\mathbf{y}\| \neq 0$. If a_i equals 0, then x_i also equals 0 using the equation $\frac{\partial f}{\partial x_i} = 0$. If $a_i \neq 0$, then we have $\frac{x_i}{a_i} = \frac{< \mathbf{b}, \mathbf{y} >}{\|\mathbf{y}\|^2}$. Similarly, either both $b_j, y_j = 0$ or $\frac{y_j}{b_j} = \frac{< \mathbf{a}, \mathbf{x} >}{\|\mathbf{x}\|^2}$. Using these relations, we have $\mathbf{x} = k\mathbf{a}$ and $\mathbf{y} = \frac{1}{k}\mathbf{b}$ for some $k \neq 0$.

To summarize, the stationary points are the $\mathbf{x}, \mathbf{y}$ that satisfy any one of the following condition:

(1) $\|\mathbf{x}\| = 0, < \mathbf{b}, \mathbf{y} > = 0$; (2) $\|\mathbf{y}\| = 0, < \mathbf{a}, \mathbf{x} > = 0$; (3) $\mathbf{x} = k\mathbf{a}$ and $\mathbf{y} = \frac{1}{k}\mathbf{b}$ for some $k \neq 0$.

(c) The optimal objective value is 0 which is achieved at any stationary point that satisfies $\mathbf{x} = k\mathbf{a}$ and $\mathbf{y} = \frac{1}{k}\mathbf{b}$ for some $k \neq 0$. We show that the other two types of stationary points in (b) are not local minimum, therefore all local minimum points satisfy the last condition in (c) and are global minimum.

Consider the stationary point $\mathbf{x} = \mathbf{0}$ and any $\mathbf{y}$ such that $< \mathbf{b}, \mathbf{y} > = 0$. We show that $f(\epsilon\mathbf{a}, \mathbf{y} + \sigma\mathbf{b}) < f(\mathbf{0}, \mathbf{y})$ for all sufficiently small σ with $\epsilon = \sigma^2 \frac{\|\mathbf{b}\|^2}{\|\mathbf{y}\|^2}$.

$$\begin{aligned} & 2f(\mathbf{0}, \mathbf{y}) - 2f(\epsilon\mathbf{a}, \mathbf{y} + \sigma\mathbf{b}) \\ &= \sum_{i,j=1}^n (a_i b_j)^2 - \sum_{i,j=1}^n (a_i b_j - \epsilon a_i (y_j + \sigma b_j))^2 \\ &= \sum_{i,j=1}^n (a_i b_j)^2 - \sum_{i,j=1}^n \left[(1 - \epsilon\sigma) a_i b_j - \epsilon a_i y_j \right]^2 \\ &= (2\epsilon\sigma - \epsilon^2\sigma^2) \sum_{i,j=1}^n (a_i b_j)^2 - \epsilon^2 \sum_{i,j=1}^n (a_i y_j)^2 + \sum_{i=1}^n \epsilon(1 - \epsilon\sigma) a_i^2 \left[\sum_{j=1}^n b_j y_j \right] \\ &= (2\epsilon\sigma - \epsilon^2\sigma^2) \|\mathbf{a}\|^2 \cdot \|\mathbf{b}\|^2 - \epsilon^2 \|\mathbf{a}\|^2 \cdot \|\mathbf{y}\|^2 + \sum_{i=1}^n \epsilon(1 - \epsilon\sigma) a_i^2 < \mathbf{b}, \mathbf{y} > \\ &= \epsilon \|\mathbf{a}\|^2 \left[(2\sigma - \epsilon\sigma^2) \|\mathbf{b}\|^2 - \epsilon \|\mathbf{y}\|^2 \right] \end{aligned}$$

If $\|\mathbf{y}\| = 0$, we have $2f(\mathbf{0}, \mathbf{y}) - 2f(\epsilon\mathbf{a}, \mathbf{y} + \sigma\mathbf{b}) > 0$ for any small enough ϵ, σ , thus the point is not a local minimum. If $\|\mathbf{y}\| \neq 0$, since $\epsilon = \sigma^2 \frac{\|\mathbf{b}\|^2}{\|\mathbf{y}\|^2}$, we have $2f(\mathbf{0}, \mathbf{y}) - 2f(\epsilon\mathbf{a}, \mathbf{y} + \sigma\mathbf{b}) = \epsilon \|\mathbf{a}\|^2 \|\mathbf{b}\|^2 [(2\sigma - \epsilon\sigma^2 - \sigma^2)]$ which is strictly greater than 0 for all sufficiently small σ . Therefore, the point is not a local minimum.

Therefore, the only local minimum points are the $\mathbf{x}$ and $\mathbf{y}$ such that $\mathbf{x} = k\mathbf{a}$ and $\mathbf{y} = \frac{1}{k}\mathbf{b}$ for some $k \neq 0$. These are all global minimum.

1. 2. 2

$$f(x) = \|x\|^{2+\beta} = (\|x\|^2)^{\frac{2+\beta}{2}}, \quad \beta \geq 0$$

$$\nabla f(x) = \left(\frac{2+\beta}{2}\right) (\|x\|^2)^{\frac{\beta}{2}} \cdot 2x = (2+\beta)x \|x\|^\beta$$

(i) Check Lipschitz assumption $\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\|, \forall x, y \in \mathbb{R}^n, \exists \text{ constant } L > 0$

$$\|\nabla f(x) - \nabla f(y)\| = (2+\beta) \|x\|^\beta - y\|y\|^\beta \leq L \|x - y\|$$

When considering $y=0$, $(2+\beta)\|x\|^\beta \leq L, \forall x \Rightarrow \text{Impossible, Lipschitz assump. fails.}$

(ii) $x^{r+1} = x^r + S d^r = x^r - S \nabla f(x) = x^r - S(2+\beta)x^r \|x^r\|^\beta = x^r(1 - S(2+\beta)\|x^r\|^\beta)$

If we desire $x^r \rightarrow x^* = 0$, then $\|x^{r+1}\| < \|x^r\|, \forall r$ must hold.

$$x^r = 0 \Rightarrow \|x^r\| = \|x^0(1 - S(2+\beta)\|x^0\|^\beta)\| < \|x^0\|$$

$$\Leftrightarrow |1 - S(2+\beta)\|x^0\|^\beta| < 1 \Leftrightarrow 0 < S(2+\beta)\|x^0\|^\beta < 2$$

This inequality has to be satisfied for S and x^0 , and then $\|x^r\| \downarrow$.

Now, we show that $x^r \rightarrow x^* = 0$.

Suppose that c is a limit value of $\|x^r\|$.

If $c=0$, then done.

$$\text{If } c > 0, \lim_{r \rightarrow \infty} \frac{\|x^{r+1}\|}{\|x^r\|} = \frac{c}{c} = 1$$

$$\text{But we know } |1 - S(2+\beta)c^\beta| < 1 \text{ since } c \leq \|x^0\|$$

$$\Rightarrow \lim_{r \rightarrow \infty} \frac{\|x^{r+1}\|}{\|x^r\|} = |1 - S(2+\beta)c^\beta| < 1, \quad \text{---} \Rightarrow c=0$$

$\Rightarrow c=0$ always holds.

$$1, 2, 3 \quad f(x) = \|x\|^{3/2} = (\|x\|^2)^{3/4}$$

$$\nabla f(x) = \frac{3}{4} (\|x\|^2)^{-1/4} \cdot 2x = \frac{3}{2} \|x\|^{-1/2} x = \frac{3}{2} \frac{x}{\|x\|^{1/2}}$$

(i) Check $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$

consider $x = -y$, $\frac{3}{2} \frac{1}{\|x\|^{1/2}} \|x - (-x)\| \leq L\|2x\| = 2L\|x\|$

$$\Rightarrow \frac{3}{2} \frac{2\|x\|}{\|x\|^{1/2}} \leq 2L\|x\| \Rightarrow 3\|x\|^{1/2} \leq 2L \Rightarrow \|x\| \leq \frac{4}{9}L^2$$

$$\Rightarrow \|x\|^{1/2} \geq \frac{3}{2L} \text{ This does NOT hold for } \forall x$$

(ii) $x^{r+1} = x^r - S \nabla f(x^r) = x^r (1 - S \frac{3}{2} \frac{1}{\|x^r\|^{1/2}})$

Desire $x^* = 0$, $\|x^{r+1}\| < \|x^r\|$, $\forall r$ must hold

$$\|x^{r+1}\| = \|x^r (1 - S \frac{3}{2} \frac{1}{\|x^r\|^{1/2}})\| \leq \|x^r\|$$

$$\Leftrightarrow |1 - S \frac{3}{2} \frac{1}{\|x^r\|^{1/2}}| < 1 \Leftrightarrow 0 < S \frac{3}{2} \frac{1}{\|x^r\|^{1/2}} < 2$$

$$\Leftrightarrow \|x^r\|^{1/2} > \frac{3}{4}S \Leftrightarrow \|x^r\| > \frac{9}{16}S^2$$

We can summarize

$$\left. \begin{array}{l} \text{If } \|x^r\| > \frac{9}{16}S^2, \text{ then } \|x^{r+1}\| < \|x^r\| \\ \text{If } \|x^r\| = \frac{9}{16}S^2, \text{ then } \|x^{r+1}\| = \|x^r\| \\ \text{If } \|x^r\| < \frac{9}{16}S^2, \text{ then } \|x^{r+1}\| > \|x^r\| \end{array} \right\} x^r \text{ never converges to } x^* = 0$$

However, if $\|x^r\| > \frac{9}{16}S^2$ for some r and then $x^{r+1} = 0$.

$\Rightarrow$ The method converges in a finite number of iterations

1.2.7

Yes, It can be viewed as steepest descent with exact line search initialized at a point with $\frac{\partial f(x_1, x_2)}{\partial x_1} = 0$.

Suppose there is an underlying function $f(x_1, x_2)$ that the engineer is trying to maximize. Assume $f(x_1, x_2)$ satisfy all assumptions for steepest descent to converge. Since after tuning x_1 first, $f(x_1, x_2)$ is maximized along x_1 direction, we have

$$\frac{\partial f(x_1, x_2)}{\partial x_1} = 0.$$

Denote the point after tuning x_1 as (x_1', x_2') , start running steepest descent with exact line search at (x_1', x_2') , we have

$$x_1^{t+1} = x_1^t + d_t \frac{\partial f(x_1^t, x_2^t)}{\partial x_1}$$

$$x_2^{t+1} = x_2^t + d_t \frac{\partial f(x_1^t, x_2^t)}{\partial x_2}$$

Since $\frac{\partial f(x_1', x_2')}{\partial x_1} = 0$, at $t=1$, we are only tuning x_2 , since we use exact line search, the function is maximized along x_2 direction and $\frac{\partial f(x_1^2, x_2^2)}{\partial x_2} = 0$, then at $t=2$, we are only tuning x_1 .

Easy to see, we will cyclically update x_1 and x_2 which is what the engineer is doing.